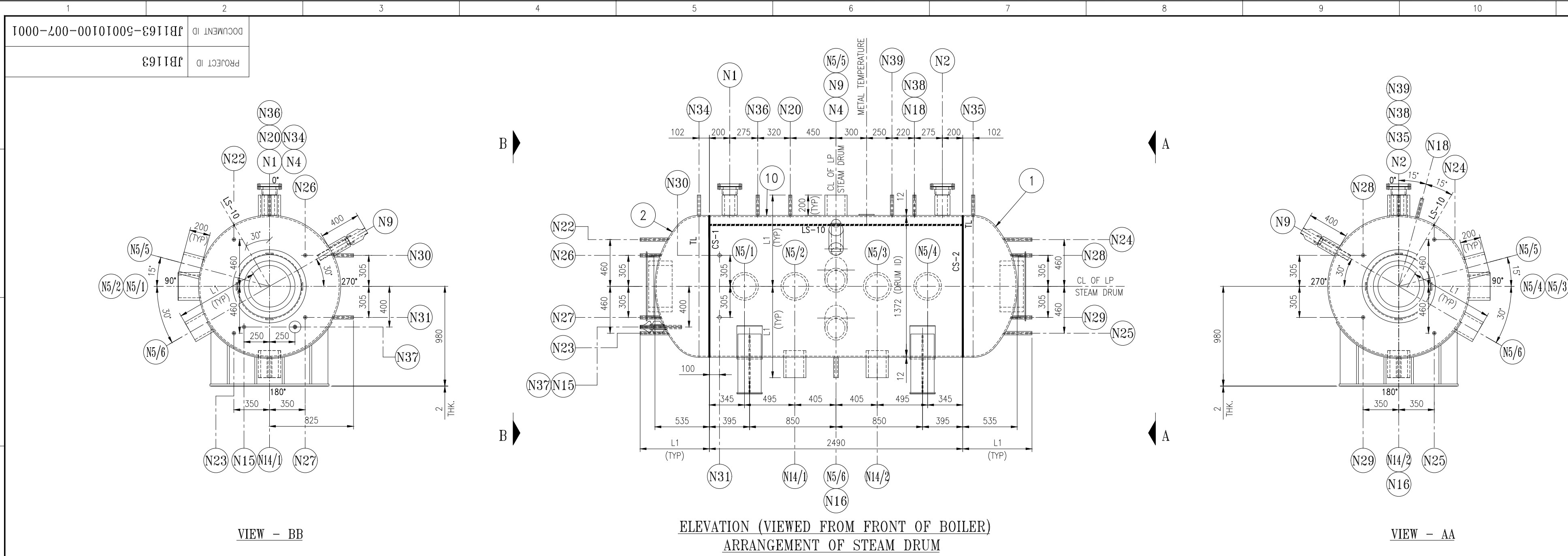
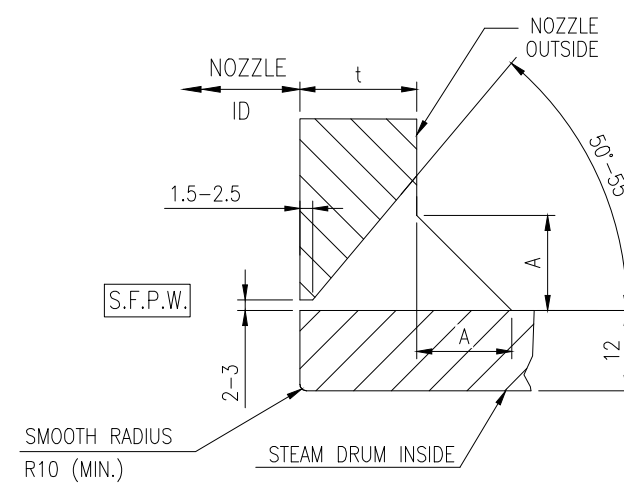


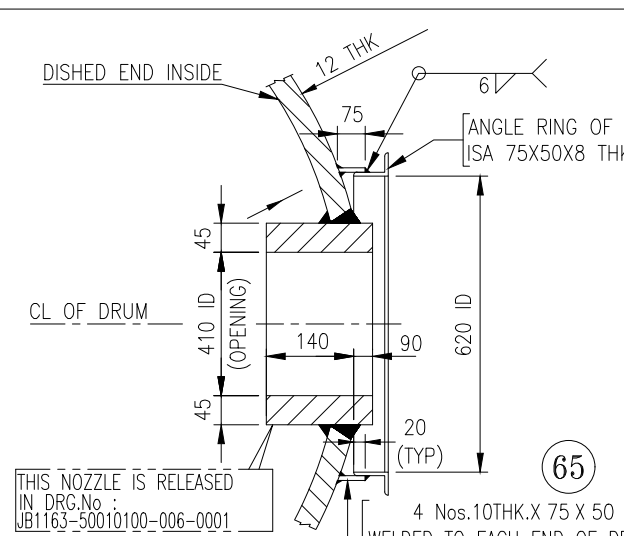


ELEVATION (VIEWED FROM FRONT OF BOILER)
ARRANGEMENT OF STEAM DRUM

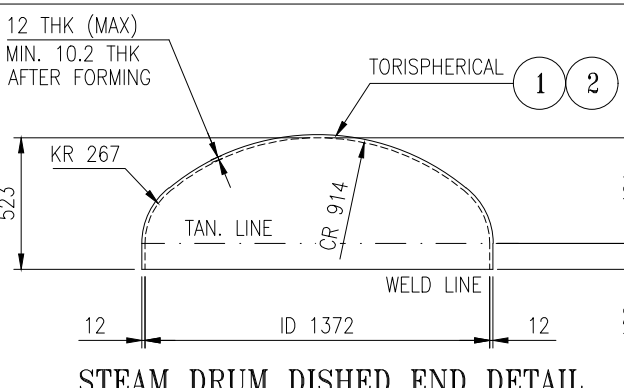
IBR CHAPTER - XII
IBR FIG : XII/85E B2



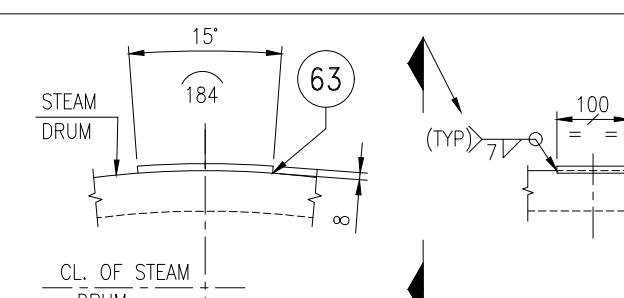
(MARK Nos. N1, N2)
WELD DETAIL OF NOZZLE TO STEAM DRUM



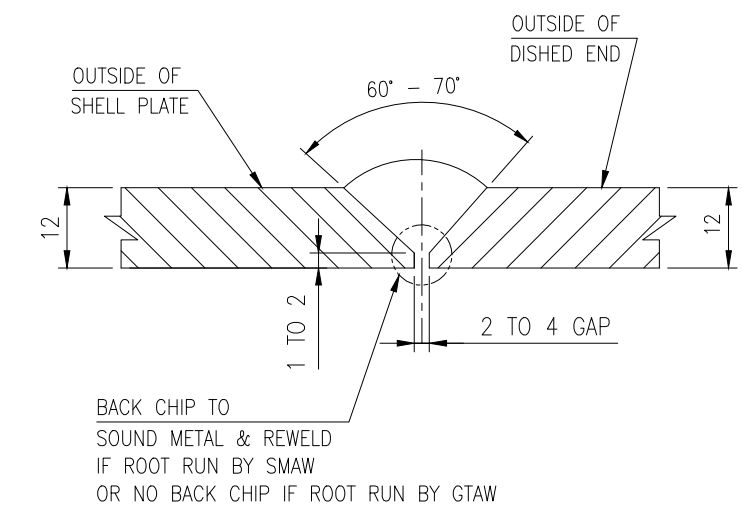
DETAIL OF MANHOLE OPENING



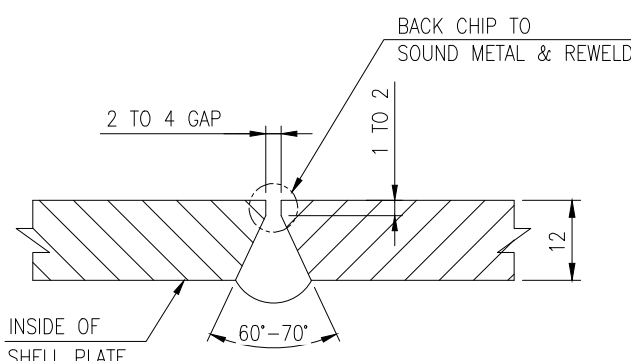
STEAM DRUM DISH END DETAIL



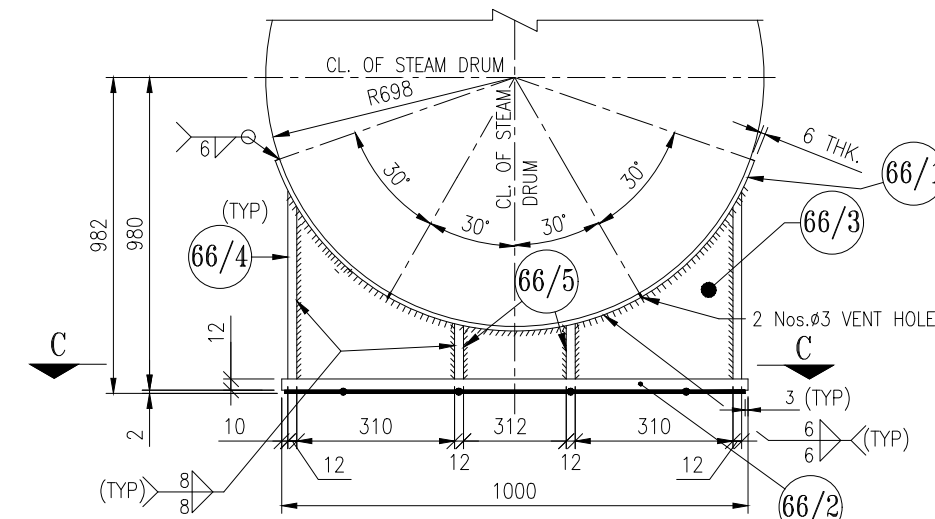
DETAIL OF METAL TEMPERATURE



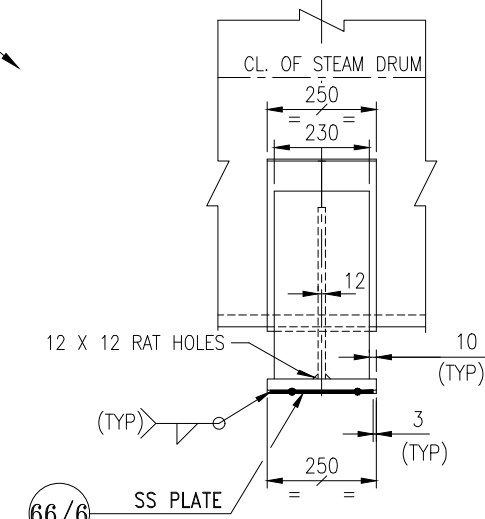
WELD DETAIL OF CLOSING SEAM FOR STEAM DRUM
(SEAM MARK Nos. CS-1 & CS-2)



WELD DETAIL OF LONGITUDINAL SEAM
FOR STEAM DRUM
(SEAM MARK Nos. LS-10)



DETAIL OF STEAM DRUM SLIDING SUPPORT (PERMANENT)



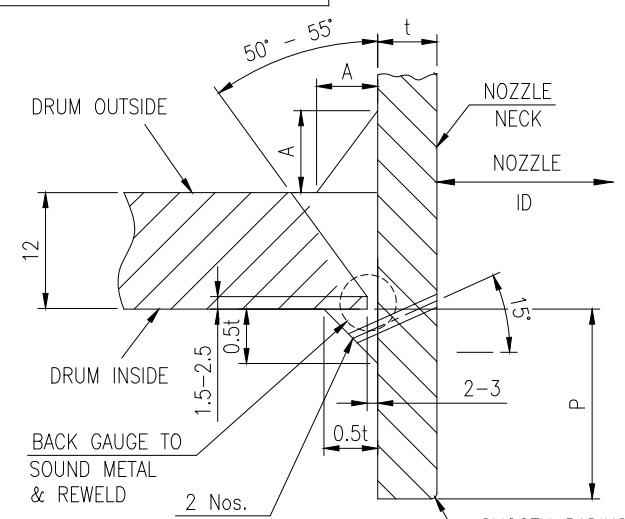
SECTION - CC

TABLE - 1									
S.Nos.	Nozzle Nos.	Nozzle Descriptions	Location of Nozzle	No. of Nozzle	Nozzle Size (OD) (mm)	Schedule of Nozzle (t) (mm)	Nozzle Thickness (t) (mm)	Connecting Pipe Schedule/Thickness (mm)	Projection from Cl. of Drum (t) (mm)
1	N1	SAFETY VALVE NOZZLE-1	STEAM DRUM SHELL	1	108	14	14	14	14
2	N2	SAFETY VALVE NOZZLE-2	STEAM DRUM SHELL	1	108	14	14	14	14
3	N4	LP SUPER HEATER STEAM STUB	STEAM DRUM SHELL	1	219.1	SCH.160	23.01	SCH.20	898
4	N5/1 TO N5/4	LP EVAPORATOR RISER STUB - I	STEAM DRUM SHELL	4	273.0	SCH.XXS	25.4	SCH.20	898
5	N5/5 TO N5/6	LP EVAPORATOR RISER STUB - II	STEAM DRUM SHELL	2	219.1	SCH.160	23.01	SCH.20	898
6	N9	FEED INLET THERMAL SLEEVE	STEAM DRUM SHELL	1	114.3	SCH.XXS	17.12	REF.009-0001	1098
7	N14/1 & N14/2	FEEDER NOZZLE	STEAM DRUM SHELL	2	219.1	SCH.160	23.01	SCH.20	898
8	N15	CBD NOZZLE	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	680
9	N16	IBD NOZZLE	STEAM DRUM SHELL	1	48.3	SCH.XXS	10.15	SCH.40	898
10	N18	AIR VENT NOZZLE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	898
11	N20	NITROGEN BLANKETING NOZZLE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	898
12	N22	LEVEL GAUGE NOZZLE (STEAM)	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	680
13	N23	LEVEL GAUGE NOZZLE (WATER)	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	680
14	N24	LEVEL GAUGE NOZZLE (STEAM)	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	680
15	N25	LEVEL GAUGE NOZZLE (WATER)	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	680
16	N26	LEVEL TRANSMITTER (STEAM)	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	680
17	N27	LEVEL TRANSMITTER NOZZLE (WATER)	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	680
18	N28	LEVEL TRANSMITTER (STEAM)	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	680
19	N29	LEVEL TRANSMITTER NOZZLE (WATER)	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	680
20	N30	LEVEL TRANSMITTER (STEAM)	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	825
21	N31	LEVEL TRANSMITTER NOZZLE (WATER)	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	825
22	N34	PRESSURE GAUGE NOZZLE (LOCAL)	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	898
23	N35	REMOTE PRESSURE GAUGE NOZZLE (FLOOR)	STEAM DRUM DISHED	1	33.4	SCH.XXS	9.09	SCH.40	898
24	N36	PRESSURE TRANSMITTER NOZZLE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	898
25	N37	HP DOSING NOZZLE WITH THERMAL SLEEVE	STEAM DRUM DISHED	1	115	32.0	REF.009-0001	680	150
26	N38	PRESSURE TRANSMITTER NOZZLE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	SCH.40	898
27	N39	SPARE NOZZLE	STEAM DRUM SHELL	1	33.4	SCH.XXS	9.09	898	9

1. HOLD HAS BEEN REMOVED FOR SAFETY VALVE NOZZLE Nos. N1 & N2 AND CORRESPONDINGLY ITS WELD DETAIL, TABLE - 1 HAS BEEN UPDATED.

V.KAVASKAR R.GAMR R.RANUNKUMAR 12.02.2020

IBR CHAPTER - XII
IBR FIG : XII/85E B1

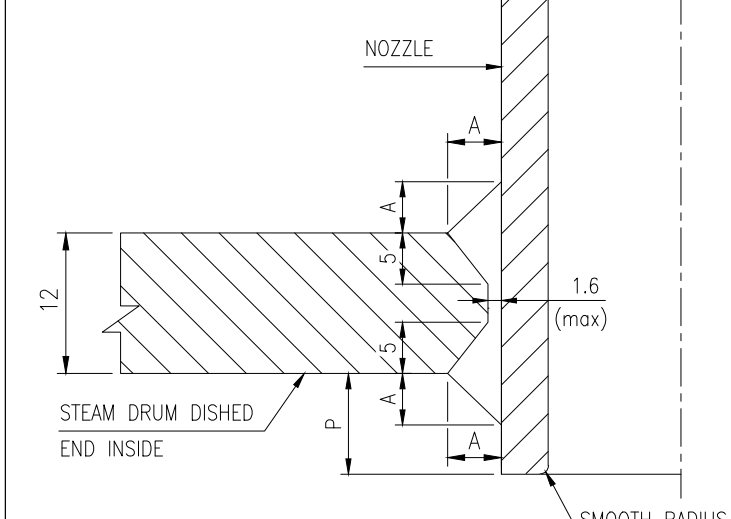


(FOR MARK Nos. N14/1 & N14/2)
WELD DETAIL OF NOZZLE TO DRUM

(MARK Nos. N4, N5/1 TO N5/6, N9, N16, N18, N20, N22, N23, N24, N25, N26, N27, N28, N29, N30, N31 N34, N35, N36, N38 & N39)

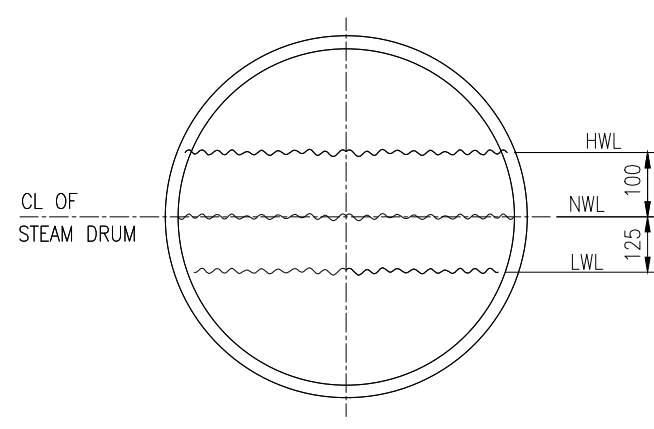
WELD DETAIL OF NOZZLE TO STEAM DRUM & DISHED END

IBR CHAPTER - VIII
IBR FIG : VIII/365/13

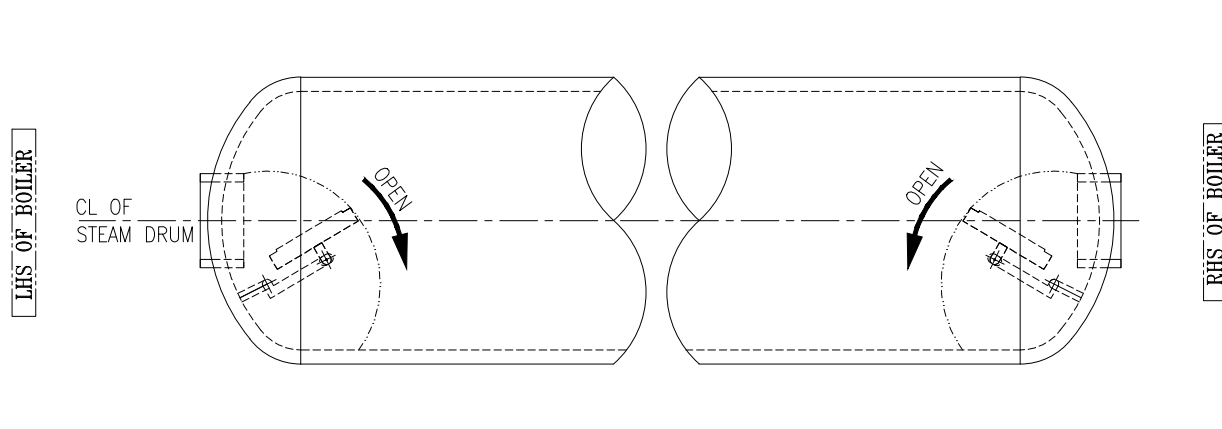


(MARK No. N15)

WELD DETAIL OF NOZZLE TO DISHED END



WATER LEVELS FOR STEAM DRUM



(FRONT OF BOILER)
MANHOLE DOOR ORIENTATION

NOTES :-

1) GENERAL :

01. THE ZERO DATE OF THIS CONTRACT IS 22.10.2019 HENCE IBR AMENDMENTS UPTO OCTOBER 2019 HAVE BEEN TAKEN CARE OF WHILE DESIGNING THE EQUIPMENT.
02. INSPECTION BY DIRECTOR OF BOILERS, UTTAR PRADESH/Lloyds CBB Authorisation, form XVA See Regulation 4C(2).
03. BECAUSE MANUFACTURING OF THIS EQUIPMENT SHALL BE DONE AT ATV PROJECT INDIA LIMITED, MATHURA, UTTAR PRADESH.
04. ALL DIMENSIONS ARE IN "mm" UNLESS OTHERWISE SPECIFIED.
05. ALL WELD JOINTS ARE 6mm UNLESS OTHERWISE SPECIFIED.
06. FOR DIMENSIONAL TOLERANCE REQUIREMENTS PLEASE REFER DOC. NO. GEN-50990100-DSM-002 AND ITS ASSOCIATED DRAWINGS.
07. LEGEND :

H.W.L.	: DENOTES HIGH WATER LEVEL.	ID	: DENOTES INSIDE DIAMETER.
N.W.L.	: DENOTES NORMAL WATER LEVEL.	SF	: DENOTES STRAIGHT FACE.
L.W.L.	: DENOTES LOW WATER LEVEL.	CR	: DENOTES CROWN RADIUS.
CS-1&CS-2	: DENOTES CLOSING SEAM.	KR	: DENOTES KNUCKLE RADIUS.
LS-10	: LONGITUDINAL SEAM.	S.F.P.W.	: SET ON FULL PENETRATION WELD.
TL	: TANGENTIAL LINE.		
LP	: LOW PRESSURE.		

II) SHOP NOTES :-

01. ALL CIRCUMFERENTIAL DIMENSIONS ARE MEASURED OUTSIDE OF DRUMS.
02. MANHOLE CIRCULAR TYPE SIZE #410 FOR DRUMS REF. DRG. No. JB1163-50010100-006-0001
03. A MAX. 45 mm FILET WELD TO BE USED FOR WELDING OF CLEAT BRACKETS AS PER DRG.No. : 011-0001 TO 014-0001 AFTER STRESS RELIEFING.
04. THIS DRAWING SHALL BE READ IN CONJUNCTION WITH THE FOLLOWING DRGS. :
 - i. FOR NOZZLE & STUB DETAILS REFER DRG.No. : JB1163-50010100-009-0001
 - ii. FOR EXTERNAL AND INTERNAL Baffles & PIPE REFER DRG.No. : 010-0001 TO 015-0001.
05. ALL WELDING TO BE DONE BY IBR QUALIFIED WELDER USING IBR WELDING PROCEDURES
06. FOR BOILER DRUMS HAVING SHELL THICKNESS 100mm AND ABOVE FOLLOWING ADDITIONAL TESTING SHALL BE CARRIED OUT :
 - i. ULTRASONIC EXAMINATION OF ALL BUTT WELD BEFORE AND AFTER PWHT.
 - ii. MAGNETIC PARTICLE EXAMINATION OF NOZZLES/STUBS TO SHELL WELDING.
08. ALL THE PAWS WELDED ALL-ROUND ON PRESSURE PART SHALL BE PROVIDED WITH 2 Nos 1/8" (16") TELL-TALE HOLES AT 180° APART ON THE PAD TO RELEASE ENTRAPPED AIR/GAS.
09. TOLERANCE FOR FORMED HEADS :-
 - THE INNER SURFACE OF A TORISPHERICAL SHALL NOT DEVIATE OUTSIDE OF THE SPECIFIED SHAPE BY MORE THAN 1/8" OF NOMINAL INSIDE DIAMETER & INSIDE THE SPECIFIED SHAPE BY MORE THAN 3/8" OF NOMINAL INSIDE DIAMETER.
10. SUITABLE STIFFENER ARRANGEMENT SHALL BE PROVIDED AT SHOP FOR NOZZLES AT DISHED END TO AVOID DAMAGE DURING TRANSPORTATION.
11. MANHOLE DOORS ARE TO BE PLACED INSIDE THE DRUM BEFORE ASSEMBLING DRUM ENDS FOR FINAL CIRCUMFERENTIAL WELD AS PER ORIENTATION DETAIL.
12. 2mm THK. SS PLATE SHALL BE PROTECTED USING WOODEN BLANKS DURING TRANSPORTATION.
13. PRIOR TO HYDRAULIC TESTING FOLLOWING SHALL BE CARRIED OUT :
 - a) ALL WELD SPATTER, WELD STUBS, SCALE, DIRT ETC SHALL BE REMOVED FROM DRUM.
14. EQUIPMENT SHALL BE THOROUGHLY DRY/CLEANED AFTER HYDRO TESTING.
15. HYDRO TESTING TO BE CARRIED OUT WITH GASKET THAT IS THE SAME AS THE SPECIFIED SERVICE GASKET.
16. HYDROTEST WATER SHALL NOT CONTAIN MORE THAN 10ppm SUSPENDED SOLIDS.

17. ALL INSIDE EDGES OF NOZZLES AND CONNECTIONS SHALL BE GROUND SMOOTH TO 3 MM RADIUS.

18. STRAY ARROD SHOULD BE AVOIDED IF ANY STRAY ARROD NOTED SHALL BE DRESSED WP & UT CHECKED.

19. FOLLOW APPROVED AND APPLICABLE NPS FOR SELECTION OF WELDING CONSUMABLES.

20. WELD SURFACES SHALL BE FREE FROM CRACKS, UNDERCUTS, UNNECESSARY GRINDING, ABRUPT RIDGES AND VALLEYS.

21. VISUAL INSPECTION OF WELD SHALL BE CARRIED OUT AND NOTED BEFORE OFFERING FOR ANY NON DESTRUCTIVE EXAMINATION.

22. HYDRO TEST WATER QUALITY SHALL BE MAINTAINED WITHIN LIMITS SPECIFIED IN DOCUMENT NO.- BLS-PPA-HYD-01-51.

23. DRUM TO BE STAMPED AS FOLLOWS :-

ITEM CODE-SDM001-LP STEAM DRUM			
ATV PROJECTS INDIA LTD., MATHURA (U.P.)			
(A/C ISGEC HEAVY ENGINEERING LTD.)			
WORKS NO -	/WIT NO. 6664-LP		
YEAR OF MAKE-			
TESTED TO -9.0 Kg/Sq. cm (g)			
TESTED ON -			
MAXIMUM WORKING PRESSURE - 6.0 Kg/Sq. cm (g)			
COMPETENT PERSON'S /			
INSPECTING AUTHORITY'S OFFICIAL STAMP			
FINAL DATE OF EXAMINATION -			

PAINTING NOTES:-

01. FOR PAINTING NOTES REFER SPECIFICATION NO:JB1163-50570900-PSH-0001.

DESIGN DATA

CODE OF CONSTRUCTION	I.B.R. (1950) & IT'S LATEST AMMENDMENTS UPTO OCTOBER 2019 (SEE NOTE : 01)
MAXIMUM WORKING PRESSURE	6.0 Kg/Sq. cm (g)
MAXIMUM WORKING TEMPERATURE	164° C
WORKING METAL TEMPERATURE	164° C
HYDRAULIC TEST PRESSURE	9.0 Kg/Sq. cm (g) AT SHOP & AS PER IBR AT SITE
RADIOGRAPHY OF LONG.&CIRCUM.SEAM	100%
STRESS RELIEVING	YES & REFER HEAT TREATMENT CYCLE TABLE
EVAPORATION OF BOILER (M.C.R.)	8.65 TPH
TOTAL HEATING SURFACE AREA OF BOILER	5295 Sq.m (APPROX.)
HEATING SURFACE AREA OF ECONOMISER (NON STEAMING)	NIL
HEATING SURFACE AREA OF DRUM	NIL (OUT SIDE GAS PATH)
STEAM PRESSURE AT MSHV OUTLET	4.0 Kg/Sq. cm (g)
STEAM TEMPERATURE AT MSHV OUTLET	195°C
WEIGHT OF DRUM	2991.05 Kgs
INSPECTION	DIRECTOR OF BOILERS, UTTAR PRADESH/Lloyds (SEE NOTE : 02)

MATERIAL SPECIFICATIONS :

DESCRIPTION	SIZE	SPECIFICATION	MIN. TENSILITY (Kg/ Sq.mm)	MIN.CALC. THK. (mm)	REMARKS
SHELL BELTS	AS PER MATERIAL LIST	SA 516 Gr.70	49.45	3.69	FOR STEAM DRUM
DISHD ENDS	AS PER MATERIAL LIST	SA 516 Gr.70	49.45	3.73	FOR STEAM DRUM

LIST OF REFERENCE DRAWINGS

DRAWING DOCUMENT ID	DESCRIPTION
JB1163-50010100-006-0001	ARRANGEMENT & DETAIL OF CIRCULAR MANHOLE ON LP STEAM DRUM FOR PH
JB1163-50010100-009-0001	DETAILS OF LP STEAM DRUM NOZZLES FOR PH

PWHT CYCLE :- (FOR LP STEAM DRUM)

- A) LOADING TEMP. (MAX.) = 300°C
- B) RATE OF HEATING ABOVE 300°C = 120°C/Hr.
- C) SOAKING TEMP. = 610±10°C
- D) SOAKING TIME (MIN.) = 30 min.
- E) RATE OF COOLING UP TO 300°C = 120°C/Hr.
- F) UNLOADING TEMP. = 300°C
- G) COOLING AFTER UNLOADING = COOLING IN STILL AIR

FLATS & BRACKETS ON DRUM	--	--	1	28.27	JB1163-50010100-013-0002
FLATS & STUDS INSIDE DRUM	--	--	1	85.26	JB1163-50010100-012-0001
DET. OF DRUM NOZZLE	--	--	1	58.39	JB1163-50010100-009-0001
A&D OF MAN HOLE DOOR	--	--	2	516.08	JB1163-50010100-006-0001
66/6	SS PLATE	SA 240 Gr. 304	PL. 2 THK - 994 X 244	2	7.62
66/5	SUPPORT PLATE	S 2062 Gr. E250 A	PL. 12 THK - 285 X 109	8	23.44
66/4	SUPPORT PLATE	S 2062 Gr. E250 A	PL. 12 THK - 463 X 230	4	40.12
66/3	SUPPORT PLATE	S 2062 Gr. E250 A	PL. 12 THK - 956 X 451	2	81.22 AS DRAWN
66/2	BASE PLATE	S 2062 Gr. E250 A	PL. 12 THK - 1000 X 250	2	47.10
66/1	SADDLE PAD	S 2062 Gr. E250 A	PL. 6 THK - 1468 X 250	2	34.58
65	CLEAT FOR RING	S 2062 Gr. E250 A	PL. 10 THK - 75 X 50	8	2.32
64	LAGGING RING	S 2062 Gr. E250 A	ISA 75 X 50 X 8THK-2105 Lg.	2	31.15
63	PAD	SA 516 Gr.70	PL. 8 THK - 100 X 185	1	1.16 FOR METAL TEMPERATURE
10	SHELL	SA 516 Gr.70	PL. 12 THK-4550 X 2500	1	990.44 (1071.5 kg-MAT.FORECAST WT.)
2	DISHD END	SA 516 Gr.70	PL. 12 THK- 1940 X 1940	1	259.57 (354.53 kg-MAT.FORECAST WT.)
1	DISHD END	SA 516 Gr.70	PL. 12 THK- 1940 X 1940	1	259.87 (354.53 kg-MAT.FORECAST WT.)

SDM001	STEAM DRUM WITH ATTACHMENTS			P	B215	1	2991.05	-		
	P.NO.	PART DESCRIPTION	SPECIFICATION	SIZE				P.QTY	P.WT(kg)	REMARK
ITEM CODE	ITEM DESCRIPTION			L.TYPE	L.GROUP	L.QTY	L.WT (Kg) (TOTAL)	REMARK		

BOILER MAKERS NUMBER I.W.T. No. : 6664-LP

I.W.T. No. : 6664-LP

CONSULTANT	--
CLIENT	M/s. ULTRATECH NATHDWARA CEMENT LIMITED, PINDWARA,SIROHI, RAJASTHAN.
DO NOT SCALE IF IN DOUBT PLEASE ASK	THIS DESIGN IS THE PROPERTY OF M/S. ISGEC HEAVY ENGINEERING LTD. AND IS ONLY ALLOWED TO BE USED BY EXPRESSED PERMISSION AND LICENSE FROM M/S. ISGEC HEAVY ENGINEERING LTD.
FOR APPROVAL	YES FOR INFORMATION NO FOR PRODUCTION YES FOR ERECTION YES
RESPONSIBLE DEPT. MECHANICAL	DRAWN BY V.KAVASKAR
CHECKED BY R.GAMIR/R.KUMARAN	APPROVED BY R.RANUNKUMAR
DATE 28.12.2019	
TOTAL WEIGHT IN kg. 2991.05	PROJECT ID JB1163
YEAR : 2019	ELEMENT ID 50010100
	SHEET SIZE : A1
	SCALE : 1 : 30
SERVICE	WHR BOILER FOR PRE HEATER PH-1 (K -STRING)
TITLE	ARRANGEMENT & DETAILS OF LP STEAM DRUM FOR PH
DRAWING DOCUMENT ID	JB1163-50010100-007-0001
REVISION	01



Isgec Heavy Engineering Ltd.
Noida 201 301 (U.P.) India